

Tayama Kirati

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EDUCATION

Tribhuwan University - Thapathali Campus

Bachelor of Engineering in Computer Engineering

2024 - 2026

Tribhuwan University - Purwanchal Campus

Bachelor of Engineering in Computer Engineering

2022 - 2024

St. Xaviers College, Maitighar

+2 Science

2019 - 2021

TECHNICAL SKILLS

Programming Languages: C/C++, Python, JS, SQL

Libraries and Tools: Numpy, Pandas, PyTorch, NodeJS, Express, SQLite/PostgreSQL, Hadoop

Theoretical Knowledge: Linear Algebra, Calculus, Probability & Statistics, DSA, OOP, OS, Software Engineering

PROJECTS

Text-to-ASL Conversion System (Minor Project)

github.com/bhattaraisushma/Snapsigns

3D Sign Language Translation Framework

Python, NLP, T5, Blender

- Designed and developed a text-to-ASL translation system to convert written text into American Sign Language animations.
- Utilized the T5 transformer model for natural language understanding and sentence to sign mapping.
- Created 3D animated ASL gestures using Blender for accurate and expressive sign visualization.
- Built an end-to-end pipeline integrating NLP processing with real-time 3D animation rendering.

Monocular SLAM and NeRF Fusion for 3D Reconstruction (Major Project)

Real-Time Neural Rendering and Localization

Python, OpenCV, SLAM, NeRF

- Built a geometric SLAM pipeline generating optimized poses and sparse point clouds from monocular video.
- Integrated Depth Anything V2 to produce detailed depth maps, for accurate 3D reconstruction.
- Established a data pipeline that bridges SLAM and NeRF by providing optimized poses and dense depth maps for neural rendering.
- Training a real-time NeRF to synthesize photorealistic, dynamic 3D scenes from the SLAM and depth outputs.

EXPERIENCE

CERTIFICATIONS AND TRAININGS
